

GITHUB ASSIGNMENT

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ASSIGNMENT ON PYTHON ND C BOTH

Assignments (10 Problems):

1. Insert an element at end of array

PYTHON

```
1 # real
2 #Insert an element at end of array
3 arr=[1,2,3,4]
4 arr.append(9)
5 print(arr)
```

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```
PS C:\Users\User\github\real> & C:/Users/User/AppData/Local/Programs/Python/Python313/python3.13t.exe c:/Users/User/
r/github/real/README.md
[1, 2, 3, 4, 9]
PS C:\Users\User\github\real>
```

C

```
9 #include<stdio.h>
10 int main(){
11     int arr[100]={1,2,3,4};
12     int n=4;
13     for(int i=n;i>2;i--){
14         arr[i]=arr[i-1];
15     }
16     arr[2]=5;
17     n++;
18
19     for(int i=0;i<n;i++)
20         printf("%d ", arr[i]);
21     return 0;
22 }
23
24
```

1 2 5 3 4

...Program finished with exit code 0
Press ENTER to exit console.

2. Find largest element in array

Python

```

1  #real
2  arr = [5, 12, 7, 9, 21, 3]
3
4  n = len(arr)
5
6  max_value = arr[0]
7
8  for i in range(1, n):
9      if arr[i] > max_value:
10         max_value = arr[i]
11
12  print("The largest element is:", max_value)
13

```

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[Running] python -u "c:\Users\User\github\real\README.md"
The largest element is: 21

C

```

9  #include <stdio.h>
10
11 int main() {
12     int arr[] = {5, 12, 7, 81, 21, 3};
13     int n = sizeof(arr) / sizeof(arr[0]);
14     int max = arr[0];
15
16     for(int i = 1; i < n; i++) {
17         if(arr[i] > max) {
18             max = arr[i];
19         }
20     }
21
22     printf("The largest element is: %d\n", max);
23
24     return 0;
25 }
26

```



The largest element is: 81

3. Find second largest element

Python

```
1 arr = [5, 12, 7, 9, 21, 3]
2
3 # Remove duplicates (optional)
4 arr = list(set(arr))
5
6 arr.sort()
7
8 second_largest = arr[-2] # second last element
9 print("Second largest element is:", second_largest)
10
11
```

PROBLEMS

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[Running] python -u "c:\Users\User\github\real\README.md"
Second largest element is: 12

[Done] exited with code=0 in 0.078 seconds

C

```
9 #include <stdio.h>
10
11 int main() {
12     int arr[] = {5, 12, 7, 9, 21, 3}, n = 6, i, j, t;
13
14     for(i = 0; i < n-1; i++)
15         for(j = i+1; j < n; j++)
16             if(arr[i] > arr[j])
17                 t = arr[i], arr[i] = arr[j], arr[j] = t;
18
19     printf("Second largest element is: %d\n", arr[n-2]);
20     return 0;
21 }
22
```

Second largest element is: 12

4. Move all zeros to end

Python

```
1 arr = [0, 1, 0, 3, 12]
2 res = [x for x in arr if x != 0] + [0] * arr.count(0)
3 print("After moving zeros:", res)
4
5
```

PROBLEMS

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[Running] python -u "c:\Users\User\github\real\README.md"

After moving zeros: [1, 3, 12, 0, 0]

[Done] exited with code=0 in 0.088 seconds

C

```
7 *****
8
9 #include <stdio.h>
10
11 int main() {
12     int a[] = {0, 1, 0, 3, 12}, n = 5, i, j = 0;
13     int res[5];
14     for(i = 0; i < n; i++) if(a[i] != 0) res[j++] = a[i];
15     while(j < n) res[j++] = 0;
16     for(i = 0; i < n; i++) printf("%d ", res[i]);
17 }
18
```



1 3 12 0 0

...Program finished with exit code 0

Press ENTER to exit console.

5. Rotate array by one

Python

```
① README.md > ...
1  arr = [1, 2, 3, 4, 5]
2  arr = [arr[-1]] + arr[:-1]
3  print("Rotated array:", arr)
4  |

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[Running] python -u "c:\Users\User\github\real\README.md"
Rotated array: [5, 1, 2, 3, 4]

[Done] exited with code=0 in 0.101 seconds
```

C

```
8
9  #include <stdio.h>
10
11 int main() {
12     int a[] = {1, 2, 3, 4, 5}, n = 5, i, last = a[n-1];
13     for(i = n-1; i > 0; i--) a[i] = a[i-1];
14     a[0] = last;
15     for(i = 0; i < n; i++) printf("%d ", a[i]);
16 }
17
18

5 1 2 3 4

...Program finished with exit code 0
Press ENTER to exit console.
```

6. Check if array is sorted

python

```
Welcome  ① README.md 1M, M  X  git-error-1753088190542

① README.md > ...
1  arr = [1, 2, 3, 4, 5]
2  print("Sorted" if arr == sorted(arr) else "Not sorted")
3  |

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[Running] python -u "c:\Users\User\github\real\README.md"
Sorted

[Done] exited with code=0 in 0.145 seconds
```

C

```
9  #include <stdio.h>
10
11 int main() {
12     int a[] = {1, 2, 3, 4, 5}, n = 5, i, sorted = 1;
13     for(i = 1; i < n; i++)
14         if(a[i] < a[i-1]) sorted = 0;
15     printf(sorted ? "Sorted\n" : "Not sorted\n");
16 }
17
18
```

Sorted

7. Reverse a string

Python

```
1  s = input("Enter a string: ")
2  print("Reversed string:", s[::-1])
3
```

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r/github/real/README.md
Enter a string: & C:/Users/User/AppData/Local/Programs/Python/Python313/python3.13t.exe c:/Users/User/github/real/README.md
Reversed string: dm.EMDAER/laer/buhtig/resU/sresU/:c exe.t31.3nohtyp/313nohtyp/nohtyp/smargorP/lacoL/ataDppA/resU/sresU/:C &
PS C:\Users\User\github\real> & C:/Users/User/AppData/Local/Programs/Python/Python313/python3.13t.exe c:/Users/User/github/real/README.md
Enter a string: Ditipriya
Reversed string: ayirpitiD
PS C:\Users\User\github\real>

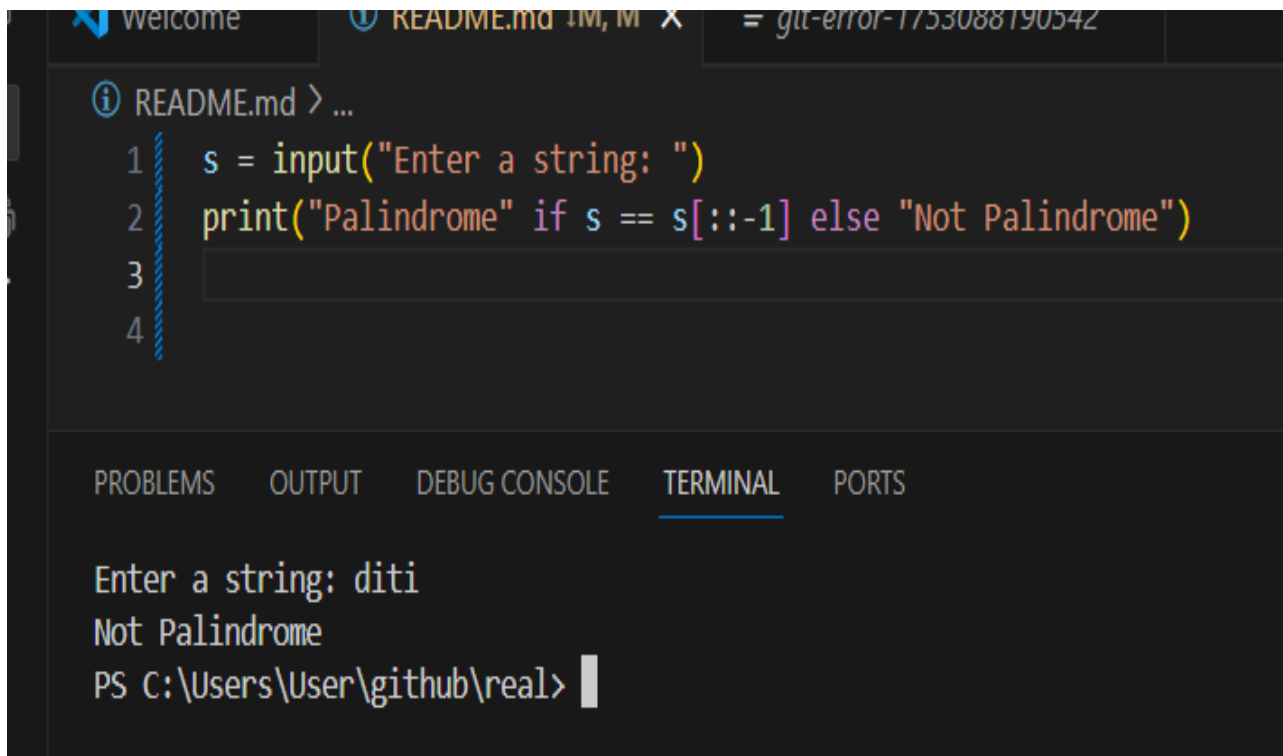
C

```
8  #include <stdio.h>
9  #include <string.h>
10
11 int main() {
12     char s[100];
13     printf("Enter a string: ");
14     scanf("%s", s); // reads a single word
15
16     int i;
17     for(i = strlen(s)-1; i >= 0; i--)
18         printf("%c", s[i]);
19     return 0;
20 }
21
```

Enter a string: ditipriya
ayirpitiD

8. Check if string is palindrome

Python

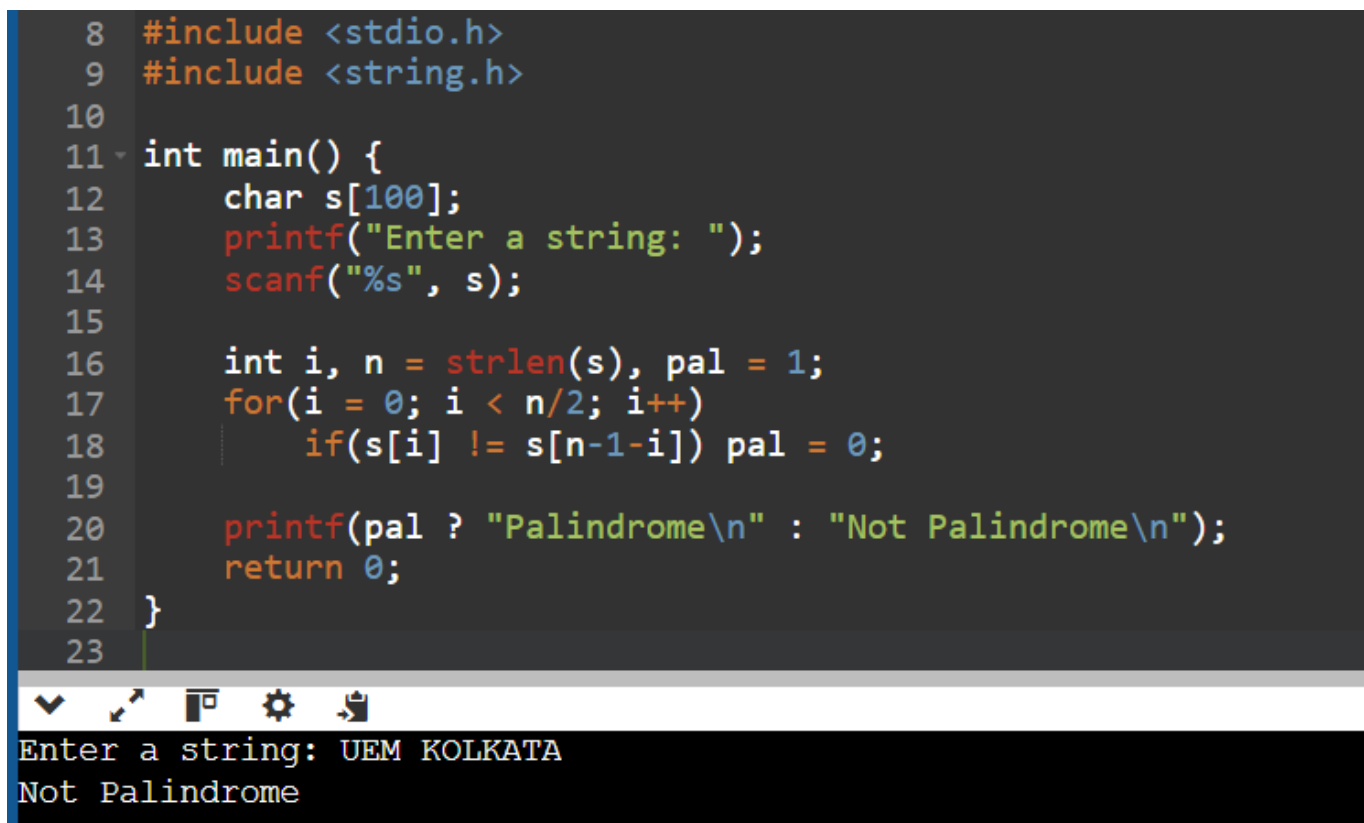


```
1 s = input("Enter a string: ")
2 print("Palindrome" if s == s[::-1] else "Not Palindrome")
3
4
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Enter a string: diti
Not Palindrome
PS C:\Users\User\github\real>

C



```
8 #include <stdio.h>
9 #include <string.h>
10
11 int main() {
12     char s[100];
13     printf("Enter a string: ");
14     scanf("%s", s);
15
16     int i, n = strlen(s), pal = 1;
17     for(i = 0; i < n/2; i++)
18         if(s[i] != s[n-1-i]) pal = 0;
19
20     printf(pal ? "Palindrome\n" : "Not Palindrome\n");
21     return 0;
22 }
23
```

Enter a string: UEM KOLKATA
Not Palindrome

9. Count frequency of array elements

Python

```
1  arr = [1, 2, 2, 3, 1, 4]
2  freq = {}
3  for num in arr:
4      freq[num] = freq.get(num, 0) + 1
5  for key, value in freq.items():
6      print(f"{key} occurs {value} times")
7
```

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```
PS C:\Users\User\github\real> & C:/Users/User/AppData/Local/Programs/Python/Python313/python3.13t.exe c:/Users/User/github/real/README.md
1 occurs 2 times
2 occurs 2 times
3 occurs 1 times
4 occurs 1 times
```

C

```
8  #include <stdio.h>
9
10 int main() {
11     int a[100], v[100] = {0}, n, i, j, c;
12
13     scanf("%d", &n);
14     for(i = 0; i < n; i++) scanf("%d", &a[i]);
15
16     for(i = 0; i < n; i++) {
17         if(v[i] == 0) {
18             c = 1;
19             for(j = i+1; j < n; j++)
20                 if(a[i] == a[j]) {
21                     c++;
22                     v[j] = 1;
23                 }
24             printf("%d occurs %d times\n", a[i], c);
25         }
26     }
27 }
28
29
```

6
8
8
9
7
6 occurs 1 times
8 occurs 2 times
9 occurs 1 times
7 occurs 1 times

----End----